AYE

• SOLAR ENERGY • LESSON 11

Battery Sizing

for Solar PV Systems

Lead-Acid • LiFePO4 • Li-NMC • DoD • BMS • Configurations

What You Will Learn

- Three battery chemistries: Lead-Acid, LiFePO4, and Li-NMC — which to choose for solar
- Key parameters: Ah, kWh, DoD, C-rate, cycle life, and temperature effects on capacity
- Battery sizing formula with step-by-step worked example for an off-grid system
- Series, parallel, and series-parallel configurations for 12V, 24V, and 48V systems

Battery Bank Sizing Formula

Rated Ah = (E_daily × Autonomy) ÷ (V_system × DoD)

E_daily = design energy (Wh)   V_system = 12/24/48V   DoD = 0.5 (Lead-Acid) or 0.8 (LiFePO4)

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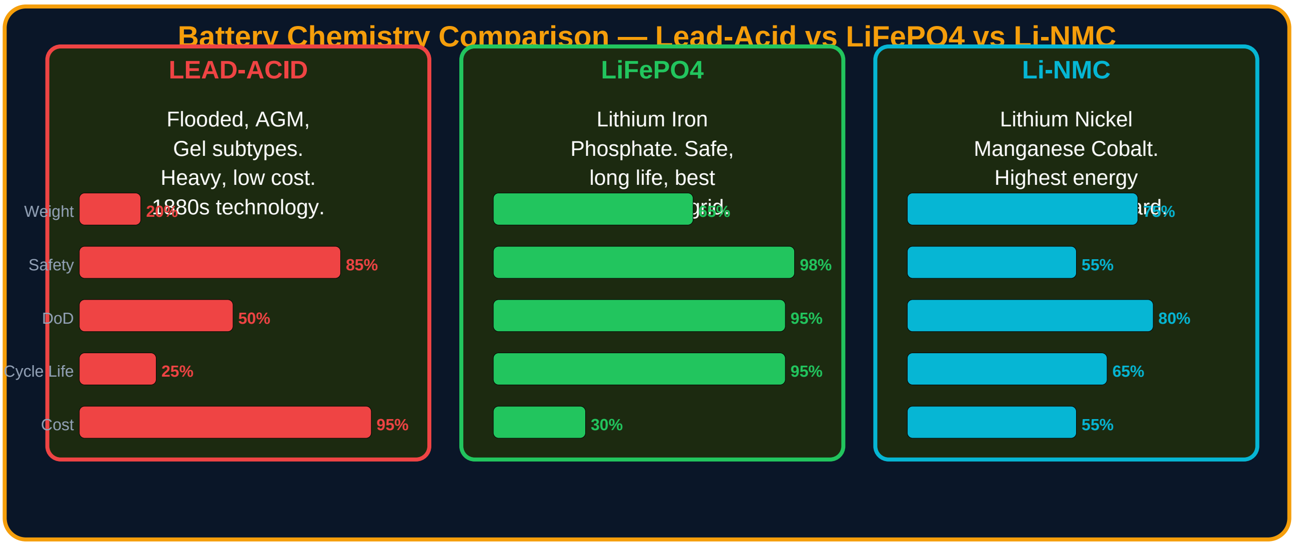
SOLAR-011: Battery Sizing for Solar PV Systems

The battery is the most expensive, most critical, and most misunderstood component in any off-grid solar system. Undersize it and the system fails on the second cloudy day. Oversize it and you waste capital that could fund more solar panels. Choose the wrong chemistry and you replace it in three years instead of fifteen. Battery sizing is where more off-grid systems fail in the design stage than anywhere else.

Learning Objectives

- Compare the three main battery chemistries (Lead-Acid, LiFePO4, Li-NMC) and select the right type for solar applications
- Define and apply key battery parameters: Ah, kWh, DoD, C-rate, cycle life, and self-discharge rate
- Apply the battery sizing formula to calculate rated Ah and kWh for any off-grid system
- Design series, parallel, and series-parallel battery configurations for 12V, 24V, and 48V systems
- Understand BMS (Battery Management System) functions and requirements

1. Battery Chemistry Comparison



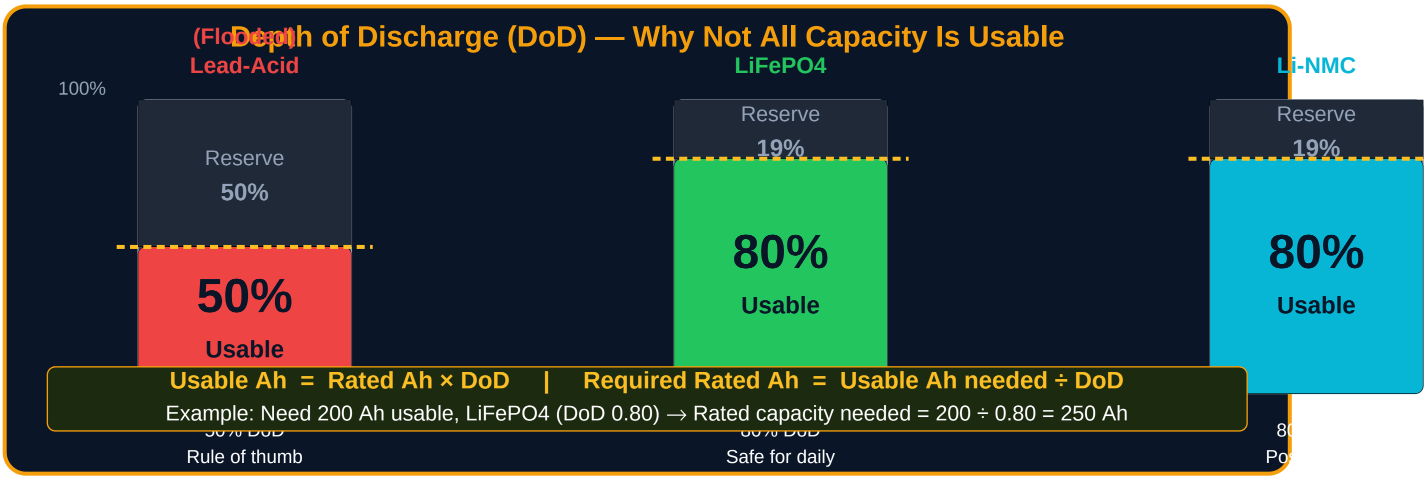
Parameter	Lead-Acid (Flooded/AGM)	LiFePO4	Li-NMC
Cycle life (80% DoD)	300–500 cycles (flooded) 500–800 (AGM)	2,000–5,000 cycles	800–2,000 cycles
DoD (recommended)	50% (flooded), 60% (AGM)	80–90%	80%

Specific energy (Wh/kg)	30–50 Wh/kg	90–160 Wh/kg	150–220 Wh/kg
Temperature range	–20°C to +50°C (charge restricted below 0°C)	–20°C to +60°C (charge limited below 0°C)	–20°C to +60°C
Self-discharge per month	3–5% per month	1–3% per month	1–2% per month
BMS required?	No (but recommended for AGM)	Yes — mandatory	Yes — mandatory
Relative cost	Lowest ($100–200/kWh)	Medium ($150–350/kWh)	Medium-high ($200–400/kWh)
Solar off-grid recommendation	Budget systems, short-term, low cycle count	**Recommended for most solar off-grid**	When weight/space critical. Higher risk if BMS fails.

2. Key Battery Parameters Every Solar Engineer Must Know

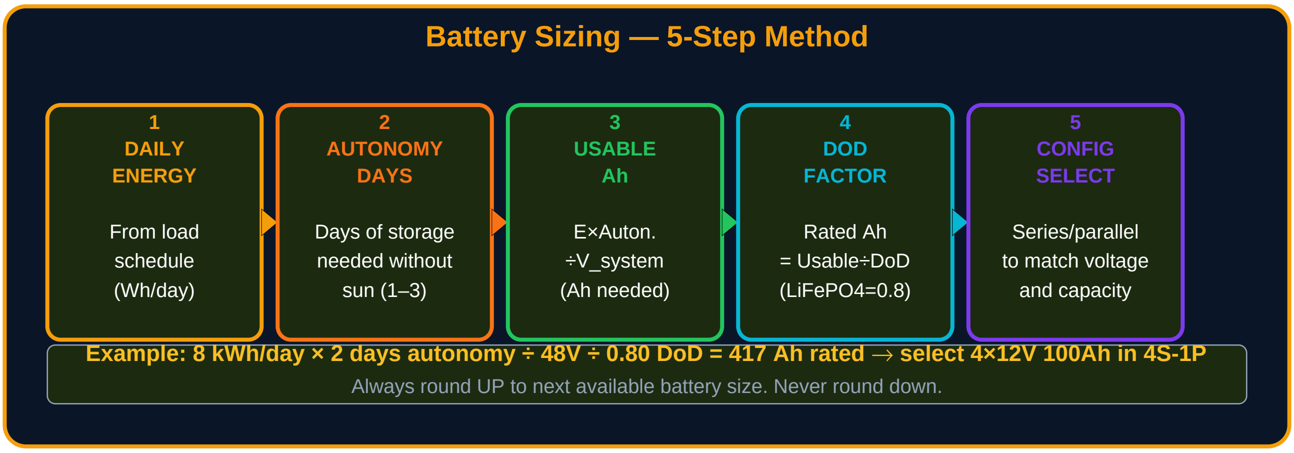
Parameter	Definition	Unit	Practical Rule
Rated Capacity	The total energy stored at full charge under standard conditions (25°C, specific discharge rate)	Ah or kWh	1 kWh = 1000 Wh. Ah × Volts = Wh. Example: 100Ah at 48V = 4,800 Wh = 4.8 kWh.
DoD (Depth of Discharge)	Maximum fraction of rated capacity that should be discharged for long service life	% or decimal	Lead-Acid: 0.50. LiFePO4: 0.80. Li-NMC: 0.80. NEVER discharge below the DoD limit repeatedly.
C-rate	The discharge or charge rate as a fraction of the rated capacity. C/20 = discharge in 20 hours.	C (dimensionless)	Solar charge controllers: typically C/10 to C/5. Avoid continuous discharge above C/5 for Lead-Acid.
Cycle life	Number of charge-discharge cycles before capacity drops to 80% of original rated value	Cycles	Lead-Acid 50% DoD: ~500 cycles. LiFePO4 80% DoD: 2,000–5,000 cycles. Use this to calculate replacement frequency.
Coulombic efficiency	Ratio of energy extracted vs energy put in during one cycle. Losses due to heat.	%	Lead-Acid: 80–85%. LiFePO4: 98–99%. LiFePO4 wastes almost no energy in charging — important for solar.
Self-discharge rate	Energy lost per month when battery is not connected to any load	% per month	Lead-Acid: 3–5%/month (significant for seasonal systems). LiFePO4: 1–3%/month.
Temperature coefficient	How much capacity changes per °C below 25°C reference temperature	% per °C	Lead-Acid: loses ~1% capacity per °C below 25°C. At 10°C: ~85% capacity. At 0°C: ~75% capacity. Design for minimum temperature on site.

3. Depth of Discharge — The Most Misunderstood Parameter



The most common battery sizing mistake is using the **full rated capacity** as the available capacity. A 200 Ah Lead-Acid battery at 50% DoD provides only 100 Ah of usable energy. An engineer who designs for 200 Ah and only gets 100 Ah will have a system that fails every night. DoD must be applied in every calculation.

4. Battery Sizing — Step by Step



Battery Sizing Formula

$$\text{Rated Battery Capacity (Ah)} = (E_{design} \times \text{Autonomy}) \div (V_{system} \times \text{DoD})$$

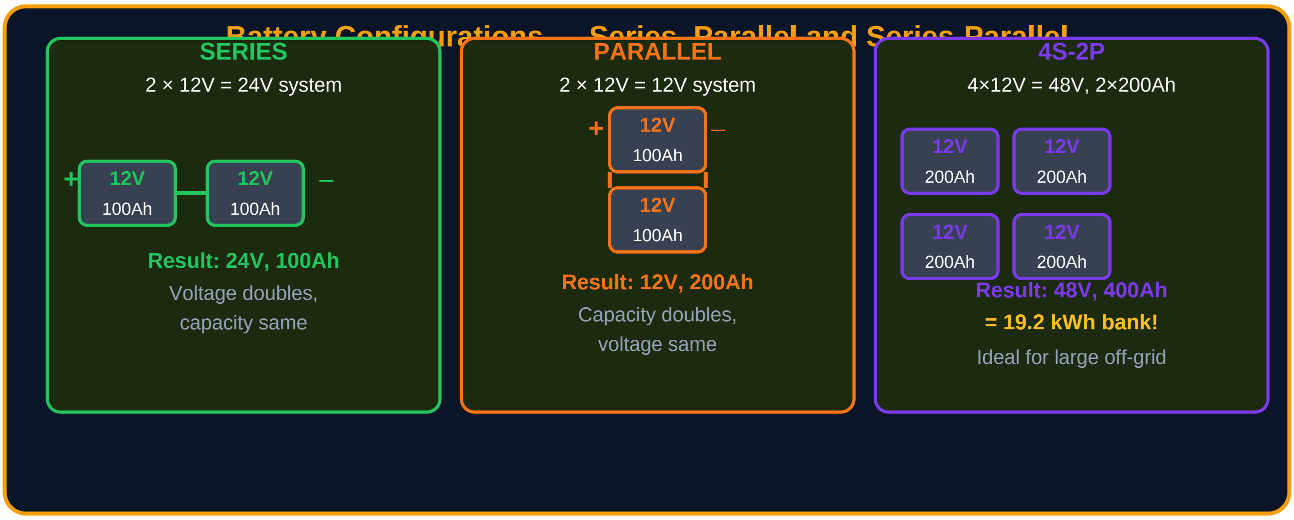
Where: E_{design} = daily design energy (Wh) from load calculation with safety margin; Autonomy = number of days of storage required (1–3 for most applications); V_{system} = system voltage (12, 24, or 48V); DoD = depth of discharge (0.5 for Lead-Acid, 0.8 for LiFePO4).

$$\text{Battery Bank in kWh} = \text{Rated Ah} \times V_{system} \div 1000$$

Worked Example — Clinic Battery Sizing

Step	Calculation	Result
Design energy from SOLAR-010	From load calculation with losses and 10% margin	8,300 Wh/day = 8.3 kWh/day
Autonomy days selected	2 days (critical health facility — 2 days without sun)	2 days
System voltage	48V DC (recommended for systems above 2 kWp)	48V
Battery chemistry selected	LiFePO4 — recommended DoD = 0.80	DoD = 0.80
Rated Ah required	(8,300 × 2) ÷ (48 × 0.80) = 16,600 ÷ 38.4	432 Ah rated
kWh capacity	432 Ah × 48V ÷ 1000	20.7 kWh
Round up to available battery	4 × 12V 120Ah batteries in 4S = 48V, 120Ah = 5.76 kWh per string → need 4 strings in parallel → 4S-4P configuration	4S-4P: 480 Ah = 23 kWh ✓

5. Battery Configurations — Series, Parallel, and Combined



6. Battery Management System (BMS)

A **Battery Management System (BMS)** is an electronic circuit board embedded in or attached to lithium battery packs that monitors and protects the cells. For LiFePO4 and Li-NMC batteries, a BMS is **not optional** — it is the single component that prevents a lithium battery from becoming a fire hazard. Never connect a lithium battery to a solar system without verifying BMS operation.

BMS Function	What It Does	Why Critical
Cell voltage monitoring	Monitors individual cell voltages and disconnects load if any cell falls below minimum (typically 2.5V for LiFePO4)	Over-discharge of a single cell causes permanent damage and potential failure of entire bank
Overcharge protection	Disconnects charger if any cell exceeds maximum voltage (typically 3.65V for LiFePO4)	Overcharging lithium cells causes thermal runaway — fire and explosion risk
Overcurrent protection	Disconnects load if current exceeds rated maximum	Prevents conductor overheating, cell damage, and fire
Temperature monitoring	Prevents charging below 0°C (lithium plating causes permanent damage) and discharging above 60°C	Lithium plating from cold charging causes internal short circuits — latent fire risk
Cell balancing	Redistributes charge among cells to keep all at equal voltage (passive or active balancing)	Unbalanced cells reduce effective capacity and cause premature failure of weakest cell
State of Charge (SoC)	Calculates and reports current state of charge as a percentage	Allows accurate battery monitoring and prevents over-discharge by load controllers

7. Temperature Effects on Battery Capacity

Battery capacity is rated at 25°C. In cold climates or when batteries are installed in uninsulated enclosures, available capacity is significantly reduced. This must be accounted for in battery sizing — particularly for systems in highland Ethiopia, East Africa, or any location with cold nights.

Temperature	Lead-Acid Available Capacity	LiFePO4 Available Capacity	Action
25°C (reference)	100% (rated capacity)	100% (rated capacity)	No derating needed
15°C	~88%	~95%	Mild derating for Lead-Acid
5°C	~75%	~85%	Increase battery bank by 15–25%
0°C	~65%	~80% (no charging below 0°C!)	Insulate batteries. Do NOT charge lithium below 0°C.
–10°C	~50%	Discharge only at reduced rate	Heated battery enclosure required

8. Practice Exercises

#	Exercise	Data
1	Chemistry selection	For each application, select Lead-Acid or LiFePO4 and justify your choice: (a) Remote weather station, budget $500, 3 kWh storage needed, replaced every 2 years. (b) Hospital off-grid system, 20 kWh storage, expected 15-year life, safety critical. (c) Temporary construction site office, 6-month use, 5 kWh storage needed.
2	DoD calculation	You need 240 Ah of **usable** energy at 48V: (a) What rated capacity (Ah) do you need with Lead-Acid (DoD = 50%)? (b) What rated capacity with LiFePO4 (DoD = 80%)? (c) Calculate the kWh capacity of each option. Which gives more energy per dollar if Lead-Acid costs $120/kWh and LiFePO4 costs $280/kWh rated?
3	Full battery sizing	Off-grid house: E_design = 5,500 Wh/day, 2 days autonomy, 24V system, LiFePO4 (DoD = 0.8). Calculate: (a) Required rated Ah, (b) kWh bank size. Available batteries: 12V 200Ah LiFePO4. Design the configuration: how many batteries, series or parallel?
4	Temperature derating	A solar system in a highland location has minimum night temperature of 5°C. System designed for 300 Ah LiFePO4 at 48V. What is the actual usable Ah at 5°C? How many Ah extra should you add to compensate?
5	BMS check	A client has purchased 200Ah LiFePO4 cells (not batteries) without BMS. They plan to connect them directly to a solar charge controller. List 4 specific risks of operating lithium cells without a BMS and explain which is the most dangerous.

Lesson Summary

Key Concept	Summary Rule
Chemistry choice	LiFePO4 for most solar off-grid — best DoD (80%), longest cycle life (2,000–5,000), safest lithium chemistry. Lead-Acid only for very tight budgets or short-term use.
DoD — never forget	Always divide required usable energy by DoD to get rated capacity. Lead-Acid: ÷0.5. LiFePO4: ÷0.8. Never design with 100% of rated capacity as usable.
Sizing formula	Rated Ah = (E_daily × Autonomy) ÷ (V_system × DoD). Autonomy = 1 day for grid-hybrid, 2–3 days for pure off-grid critical loads.
Series connections	Batteries in series → voltage adds, capacity stays the same. Two 12V 100Ah in series = 24V 100Ah.
Parallel connections	Batteries in parallel → capacity adds, voltage stays the same. Two 12V 100Ah in parallel = 12V 200Ah.
Series-Parallel	For large systems: series first to reach system voltage, then parallel to increase capacity. Always use identical batteries in each configuration.
BMS — mandatory for lithium	Never connect a lithium battery (LiFePO4 or Li-NMC) without a BMS. Over-discharge, overcharge, and cold charging cause permanent damage and fire risk.
Temperature derating	Add 15–25% battery capacity for systems at 0–10°C. Never charge lithium batteries below 0°C. Insulate battery enclosures in cold climates.

What Is Next — SOLAR-012

SOLAR-012: Inverter Sizing — The complete guide to sizing and selecting inverters for off-grid solar systems. Pure sine wave vs modified sine wave. Continuous power vs surge rating. Efficiency curves and standby consumption. Hybrid inverter-chargers (Victron MultiPlus, SMA Sunny Island). Transfer switch, AC coupling, and generator integration.

Resource	Details
SOLAR PDF Series	SOLAR-001 through SOLAR-030 — ayetechub.com/pdfs.html
Solar PV Calculator	Free interactive tool — ayetechub.com/downloads/solar-calculator.html
Community	Telegram: t.me/ayetechub